

Lecture 25

Eigenvalue IV and Singular Value I

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Outline and Review

3 mins



Roadmap of MATH2041

Segment 1

Solving linear systems (Lec 3-15);
Solving least squares problem (Lect 16,17)

Segment 2

[Lec 18-21]: Two relatively independent parts:
—Determinant. [Important tool!]
—Linear transformation. [Advanced math perspective of matrix]

Segment 3

Next: Lec 22-27: Eigenvalues and related.

Schedule:

- Eigenvalues. Lec 22-24
- Singular values. Lec 25-26
- Quadratic forms. Lec 27.

Today's Lecture

Outline today

- 1) Some properties of Eigenvalues.
- 2) Decomposition to Rank-1 Matrices.
- 3) Low-rank matrix and **data compression**
- 4) **Compact SVD**.

After today's lecture, you should be able to:

- i) Describe a few properties of eigenvalues;
- ii) Write a matrix in terms of rank-1 matrices;
- iii) Write the compact SVD;
- iii) Explain how SVD is used in data compression

Eigenvalues and Eigenvectors

Definition 21.1 (Eigenvalues and Eigenvectors)

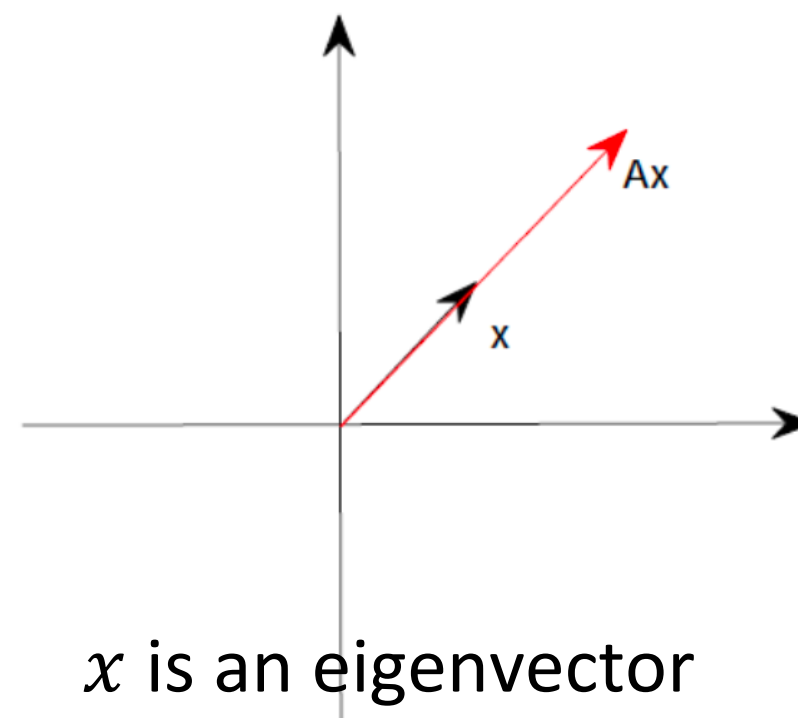
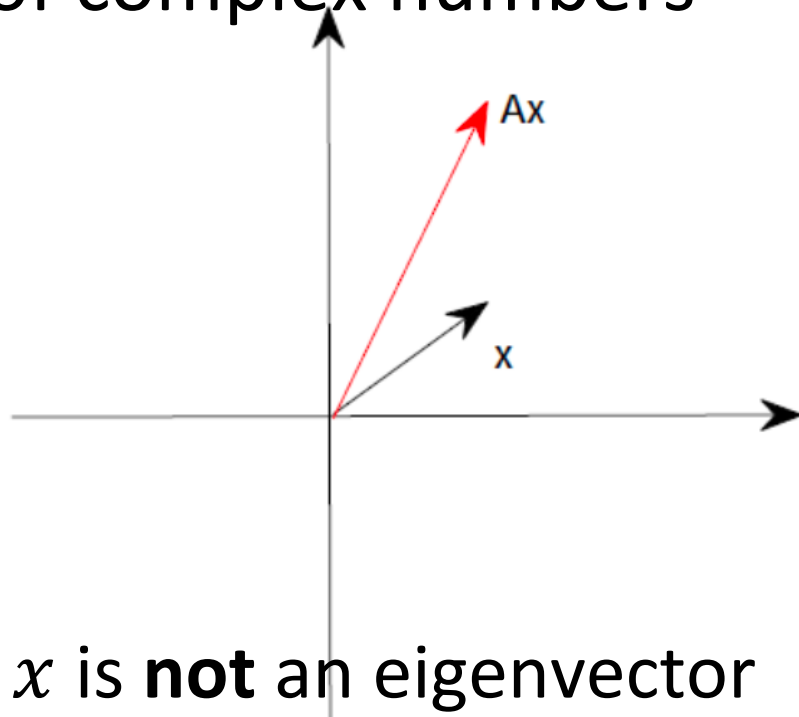
Let $A \in \mathbb{C}^{n \times n}$ be a square matrix.

If there exists a scalar λ ($\in \mathbb{R}$ or $\mathbb{C}$) and a **nonzero** vector x such that

$$Ax = \lambda x,$$

then λ is called a (real or complex) **eigenvalue** and x is called an **eigenvector** with respect to (or associated with; corresponding to) λ .

$\mathbb{C} :=$ set of complex numbers



General Procedure to Find Eigenvalues/Eigenvectors

Step 1: Solve $\det(A - \lambda I) = 0$ and get n roots $\lambda_1, \dots, \lambda_n$
Solving single-var polynomial.

Step 2: For each λ_i , find the eigenspace $N(\lambda_i I - A)$
Solving up to n linear systems.

Any nonzero vector in $N(\lambda_i I - A)$ is an eigenvector corresponding to λ_i

$$N(\lambda_i I - A) = \left\{ \begin{array}{l} \text{eigenvectors of } A \text{ with} \\ \text{respect to the eigenvalue } \lambda_i \end{array} \right\} \cup \{0\}$$

Spectral Theorem

Theorem 23.1 [Spectral Theorem]

Any real symmetric matrix A can be written as

$$A = VDV^T = \sum_{j=1}^n \lambda_j \mathbf{v}_j \mathbf{v}_j^T. \quad (*)$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a real diagonal matrix,

$V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ is a real orthogonal matrix.

(*): Eigenvalue decomposition (EVD) or eigendecomposition of A .

Diagonalization

Definition 24.2 (Diagonalizable 可对角化)

If a square matrix A is similar to a diagonal matrix, i.e., $A = SDS^{-1}$ for some diagonal matrix D , then we say A is diagonalizable.

Theorem 24.1 [Suffi. & Necc. Conditions for Diagonalizable: n indep. eigenvectors]

An $n \times n$ matrix A is diagonalizable $\Leftrightarrow$ eigenvectors of A can form a basis.

Fact: Not all square matrices are diagonalizable.

Proposition 24.2 (distinct eigenvalues $\rightarrow$ diagonalizable)

Suppose an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable.

Part I Properties of Eigenvalues

12 mins

Sum and Product of Eigenvalues

Question: What is the sum of eigenvalues?

Property 25.1 (Sum of Eigenvalues)

Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$) be a square matrix.

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii}$$

Definition: $\sum_{i=1}^n a_{ii}$ is called the **trace** of A , denoted by $\text{tr}(A)$ or $\text{Tr}(A)$.

tr

Question: What is the product of eigenvalues?

Property 25.2 (Product of Eigenvalues)

Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$) be a square matrix.

invertibility condition

$$\prod_{i=1}^n \lambda_i = \det(A)$$

$A^{-1} \nexists \Leftrightarrow \det(A) = 0 \Leftrightarrow \text{some } \lambda_i = 0 \Leftrightarrow A \cdot v = 0 \cdot v = 0$
for some nonzero v .

Example of Sum and Product of Eigenvalues

Example

$$A = \begin{bmatrix} 5 & -18 \\ 1 & -1 \end{bmatrix}$$

$5 \times (-1) - 1 \times (-18)$
 $\det(A) = 13$ and $5 + (-1)$
 $\operatorname{tr}(A) = 4.$

$$p_A(\lambda) = \det(\lambda I - A) = \begin{vmatrix} 5 - \lambda & -18 \\ 1 & -1 - \lambda \end{vmatrix} = \lambda^2 - 4\lambda + 13$$

$$\lambda_1 = 2 - 3i, \lambda_2 = 2 + 3i.$$

$$\det(A) = \lambda_1 \lambda_2 = 4 + 9 = 13, \quad \operatorname{tr}(A) = \lambda_1 + \lambda_2 = 4.$$

Conclusion: Prop 25.1 and 25.2 hold for this example.

Eigenvalues of Matrix Power and Inverse

$\text{EIG}(A)$ is the **multiset** of eigenvalues of matrix A .

Does $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ “have the same eigenvalues”?

No, since $\text{EIG}(A) \neq \text{EIG}(B)$.

Property 25.3 (Eigenvalues of Matrix Power)

Let $A \in \mathbb{C}^{n \times n}$ be a square matrix, k is a positive integer.

Suppose $\text{EIG}(A) = \#\{\lambda_1, \lambda_2, \dots, \lambda_n\}$.

Then (a) $\text{EIG}(A^k) = \#\{\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k\}$.

(b) If A^{-1} exists, then $\text{EIG}(A^{-1}) = \#\{\lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_n^{-1}\}$.

Eigenvalues of Matrix Power and Inverse

If A is diagonalizable, the proofs are easy.

(a):

$$A = SDS^{-1},$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, S is invertible.

for any k , $A^k = SDS^k S^{-1}$, where $D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$
 $\text{EIG}(A^k) = \{\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k\}.$

(b) A is invertible, so all $\lambda_i \neq 0$. $AB = (S D S^{-1})(S D^{-1} S^{-1}) = I.$
 $BA = I.$

Let $B = S D^{-1} S^{-1}$, then $AB = BA = I$.

So $B = A^{-1}$. $\text{EIG}(A^{-1}) = \{\lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_n^{-1}\}.$

full proof of (b): left to assignment 6.

$$A \sim B$$
$$\text{EIG}(A) = \text{EIG}(B)$$

Part II Preparation: Decomposition by Rank-1 Matrices

18 mins

Rank-1 Matrix

Fact: Any rank-1 matrix can be written as uv^T

e.g.

$$\underbrace{\begin{bmatrix} 1 & 0 & 3 \\ 2 & 0 & 6 \\ 4 & 0 & 12 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}}_u \underbrace{\begin{bmatrix} 1 & 0 & 3 \end{bmatrix}}_{v^T}$$

(Handwritten blue arrows: one from the first row of A to the first row of v^T, and another from the third column of A to the third column of v^T, with labels '0' and '3' respectively.)

Analysis: Suppose $A = [\mathbf{a}_1, \dots, \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ has rank 1.

Since $\dim(C(A)) = 1$, we have $\mathbf{a}_i = y_i \mathbf{x}$ for some vector $\mathbf{x} \in \mathbb{R}^{m \times 1}$.

$$A = \underbrace{[\mathbf{a}_1, \dots, \mathbf{a}_n]}_{m \times n} = \underbrace{[y_1 \mathbf{x}, \dots, y_n \mathbf{x}]}_{m \times n} = \underbrace{\mathbf{x}}_{m \times 1} \underbrace{[y_1, y_2, \dots, y_n]}_{1 \times n} = \boxed{\mathbf{x} y^T}.$$

(Handwritten blue annotations: 'm x n' under A, 'm x 1' under each a_i, '1 x 1' under y_i, 'm x 1' under x, '1 x n' under the row vector [y_1, ..., y_n], and 'm x 1' and '1 x n' under the final boxed expression x y^T.)

(Handwritten blue text: 'scalar anywhere' with an arrow pointing to the y_i term.)

$$\text{Easier way: } A = \underbrace{[\underbrace{\mathbf{x} y_1}_{m \times 1}, \dots, \underbrace{\mathbf{x} y_n}_{m \times 1}]}_{m \times n} = \underbrace{\mathbf{x}}_{m \times 1} \underbrace{[y_1, \dots, y_n]}_{1 \times n} = \boxed{\mathbf{x} y^T}$$

Rank-2 Matrices

[important exercise]

Claim: Any rank-2 matrix can be written as the sum of two rank-1 matrices.

Analysis: Suppose $A = [\mathbf{a}_1, \dots, \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ has rank 2.

Suppose $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^{m \times 1}$ form a basis of $C(A)$.

Then every column $\mathbf{a}_i = \underline{y_{i1}} \mathbf{x}_1 + \underline{y_{i2}} \mathbf{x}_2 =$

$\boxed{m \times 1} \quad 1 \times 1 \quad \boxed{m \times 1}$

$\begin{bmatrix} \boxed{x_1} & \boxed{x_2} \end{bmatrix} \begin{bmatrix} y_{i1} \\ y_{i2} \end{bmatrix}$

$\begin{matrix} \text{block-dim } 1 \times 2 & 2 \times 1 \\ m \times 1 & m \times 1 \end{matrix}$
 $\text{whole-dim } \boxed{m \times 2} \quad \boxed{2 \times 1}$

$y_{i1} \quad y_{i2} \quad \begin{matrix} x_1 \rightarrow m \times 1 \\ x_2 \rightarrow m \times 1 \end{matrix}$
 $2 \quad \textcircled{1} \quad \textcircled{2m}$

easier to understand:

$$= \underbrace{x_1}_{m \times 1} \underbrace{y_{i1}}_{1 \times 1} + \underbrace{x_2}_{m \times 1} \underbrace{y_{i2}}_{1 \times 1} = [x_1, x_2] \begin{bmatrix} y_{i1} \\ y_{i2} \end{bmatrix}$$

$\underbrace{\quad}_{i=1} \quad \underbrace{\quad}_{i=1}$
 match match

Rank-2 Matrices

Claim: Any rank-2 matrix can be written as the sum of two rank-1 matrices.

Analysis: Suppose $A = [\mathbf{a}_1, \dots, \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ has rank 2.

Suppose $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^{m \times 1}$ form a basis of $C(A)$.

Then every column $\mathbf{a}_i = y_{i1}\mathbf{x}_1 + y_{i2}\mathbf{x}_2 = [\mathbf{x}_1, \mathbf{x}_2] \begin{bmatrix} y_{i1} \\ y_{i2} \end{bmatrix}$

$m \times 1 \quad 1 \times 1 \quad m \times 1 \qquad m \times 2 \quad 2 \times 1$

$$\text{Then } A = [\mathbf{a}_1, \dots, \mathbf{a}_n] = [\mathbf{x}_1, \mathbf{x}_2] \begin{bmatrix} y_{11} & y_{21} & \dots & y_{n1} \\ y_{12} & y_{22} & \dots & y_{n2} \end{bmatrix}$$

$m \times n \quad m \times 1 \quad m \times 1 \quad m \times 2 \quad 2 \times 1 \quad 2 \times 1 \quad 2 \times 1$

$$= [\mathbf{x}_1, \mathbf{x}_2] \begin{bmatrix} y_1^T \\ y_2^T \end{bmatrix} = \mathbf{X} \mathbf{Y}^T$$

$m \times 2 \quad 2 \times n \quad m \times 2 \quad 2 \times n$

$$= \mathbf{x}_1 y_1^T + \mathbf{x}_2 y_2^T$$

$m \times 1 \quad 1 \times n \quad m \times 1 \quad 1 \times n$

$x_i y_i$

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$$

$$\mathbf{Y}^T = \begin{bmatrix} y_1^T \\ y_2^T \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \\ \vdots & \vdots \\ y_{n1} & y_{n2} \end{bmatrix}$$

Rank-r Matrices

Proposition 25.1 Suppose $A = [\mathbf{a}_1, \dots, \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ has rank r . Then

i) $\mathbf{A} = \sum_{j=1}^r \mathbf{x}_j \mathbf{y}_j^\top$,

where $\mathbf{x}'_j$ s are linearly independent, $\mathbf{y}'_j$ s are linearly independent.

ii) $\mathbf{A} = \mathbf{X}\mathbf{Y}^\top$,

where $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_r] \in \mathbb{R}^{m \times r}$, $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_r] \in \mathbb{R}^{n \times r}$ have rank r .

iii) $\text{Row}(\mathbf{A}) = \text{Row}(\mathbf{Y}^\top) = \text{span}\{\mathbf{y}_1, \dots, \mathbf{y}_r\}$,

$$\text{C}(\mathbf{A}) = \text{C}(\mathbf{X}) = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_r\},$$

Part III Motivation: Data Compression

Sec. 7.1

7 mins

Reading: SVD and Data Compression Using Low-rank Matrix Approximation
<https://dustinstansbury.github.io/theclevermachine/svd-data-compression>

Data Matrix

Data matrix can be huge.



Figure?

Human likes!

157	153	174	168	150	152	129	151	172	161	155	166
155	182	163	74	75	62	33	17	110	210	180	164
180	180	50	14	34	6	10	33	48	106	169	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234

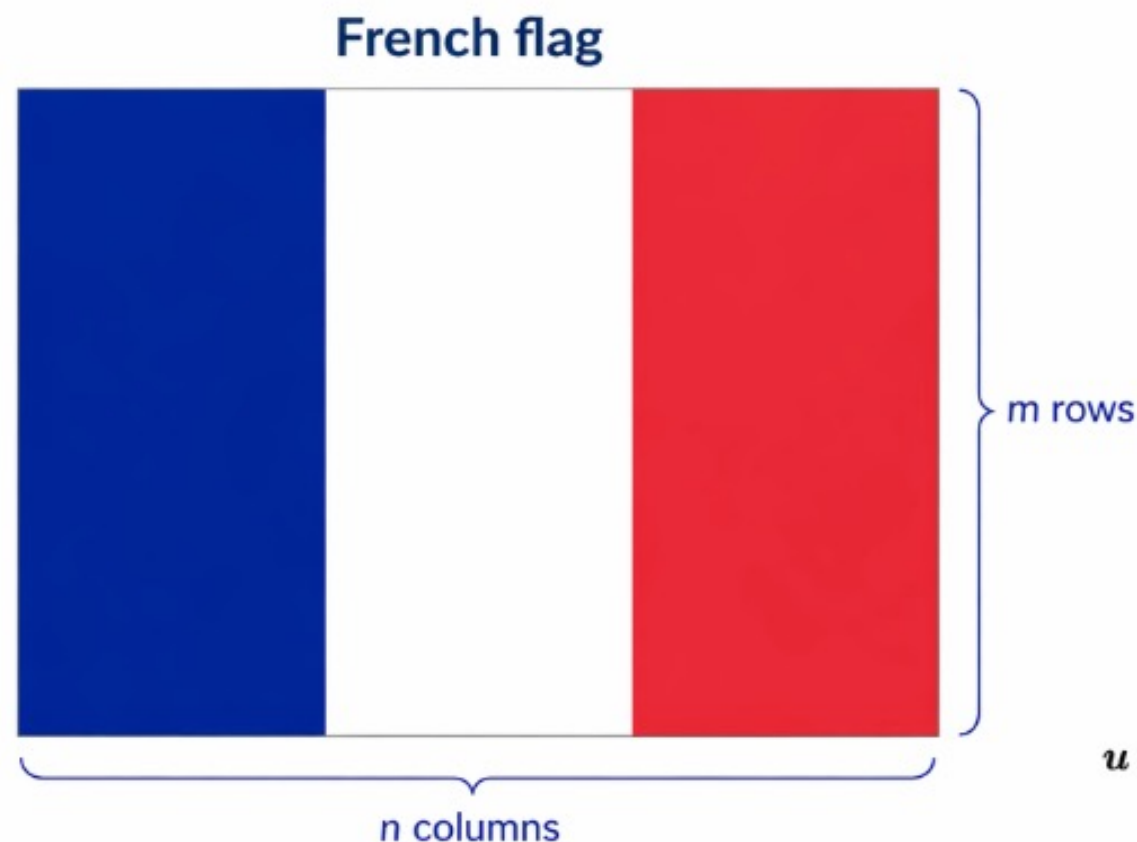
Matrix!

Computer likes!

Pictures from Lecture 1.

National Flag As Image

A flag image can be viewed as a matrix of pixel values.
Many flags have strong repeated structure.



Naive representation: send the full matrix A

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 & 2 & 2 & \cdots & 2 \\ 0 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 & 2 & 2 & \cdots & 2 \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 & 2 & 2 & \cdots & 2 \end{bmatrix} \quad (m \times n)$$

We would need to send **all** $m \times n$ entries.

Compressed idea: exploit the structure in A

Each column is constant (only three distinct columns).
So A has rank 1 and can be written as $A = uv^T$.

$$u = \begin{bmatrix} / \\ / \\ / \\ \vdots \\ / \end{bmatrix} \quad (m \times 1), \quad v^T = \begin{bmatrix} 0 & \cdots & 0 & 1 & \cdots & 1 & 2 & \cdots & 2 \end{bmatrix} \quad (1 \times n)$$

Instead of sending $m \times n$ numbers,
send only $m+n$ numbers (the vectors u and v^T).

If $m=n = 1024$,
--Naïve: storage $mn \sim 1\text{M}$;
--Rank-1: storage $m+n \sim 2\text{K}$.

Compression of Data Matrix: Exact Low Rank

Suppose A is $m \times n$ matrix.

To store A directly, we use mn numbers.

To store $A = \sum_{j=1}^r x_j y_j^T$, we use $\sim r(n+m)$ numbers.

Low rank matrix:

If $\text{rank}(A) \ll \min(m, n)$, then we say A is low rank.

$\downarrow$
much smaller

e.g. $m = n = 10^6$.

n^2 numbers require memory 7450 G.

If $r = 100$, then $2rn$ numbers require memory 1.5G.

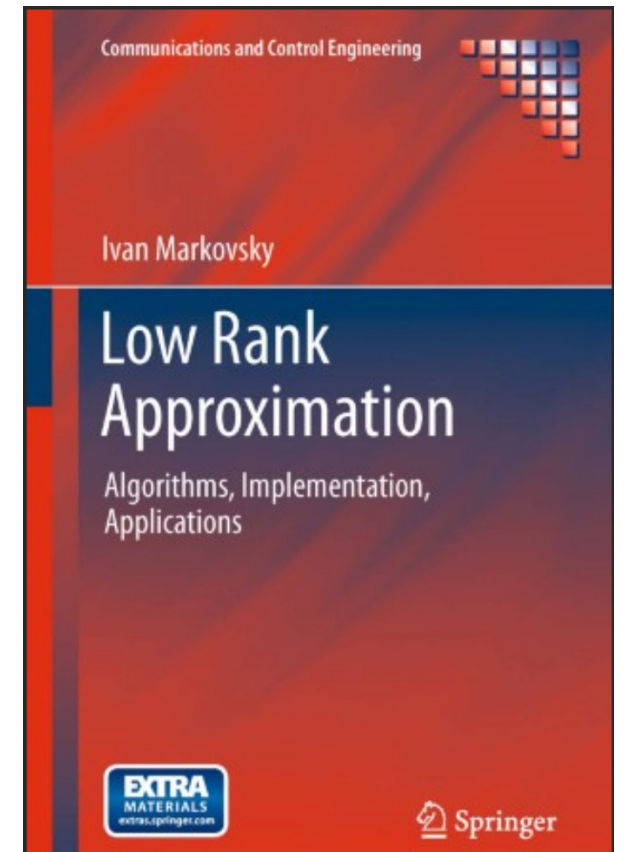
What if NOT Exact Low Rank?

In practice, exact low rank matrices are rare.

Are the discussions above **vacuous**?

No.

Practical matrices can be
“**approximately** low rank”!



Example: $A = \mathbf{u}_1 \mathbf{v}_1^T + 0.01 \mathbf{u}_2 \mathbf{v}_2^T$, $\|\mathbf{u}_i\| = \|\mathbf{v}_i\| = 1$.

Then A is “approximately rank-1”.

$\hat{A} = \mathbf{u}_1 \mathbf{v}_1^T$ is a good approximation of A .

Will make this claim more rigorous later.

Part IV SVD Theorem

8 mins

Compact SVD

Theorem 25.1 [Compact SVD and vector form]

Any matrix $A \in \mathbb{R}^{m \times n}$ with rank r can be written as

$$A = \sum_{j=1}^r \overset{1 \times 1}{\sigma_j} \overset{m \times 1}{\mathbf{u}_j} \overset{1 \times n}{\mathbf{v}_j^T} = U_r \Sigma_r V_r^T, \quad (*)$$

$\|\mathbf{u}_j\|=1=\|\mathbf{v}_j\|$

Where $U_r = [\mathbf{u}_1, \dots, \mathbf{u}_r] \in \mathbb{R}^{m \times r}$ has orthogonal unit-norm columns,
 $V_r = [\mathbf{v}_1, \dots, \mathbf{v}_r] \in \mathbb{R}^{n \times r}$ has orthogonal unit-norm columns,
 $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}_+^{r \times r}$ is a diagonal matrix.

Here $\mathbb{R}_+$ is the set of non-negative real numbers.

Remark:

By Prop. 25.1, rank- r matrix $A = \sum_{j=1}^r \mathbf{x}_j \mathbf{y}_j^T = \sum_{j=1}^r \sigma_j \hat{\mathbf{x}}_j \hat{\mathbf{y}}_j^T$, where $\hat{\mathbf{x}}_j, \hat{\mathbf{y}}_j$ have unit norm.
which has a similar form to (*);
 $\|\mathbf{x}_j\| \cdot \|\mathbf{y}_j\|, \quad \hat{\mathbf{x}}_j = \frac{\mathbf{x}_j}{\|\mathbf{x}_j\|}, \hat{\mathbf{y}}_j = \frac{\mathbf{y}_j}{\|\mathbf{y}_j\|}$

but there is one more requirement: we want

- $\mathbf{u}_1, \dots, \mathbf{u}_r$ are orthogonal to each other;
- $\mathbf{v}_1, \dots, \mathbf{v}_r$ are orthogonal to each other;

Proof is nontrivial; skip.

Compact SVD Examples

Example 1 (square 3×3 matrix, rank 2):

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}_{3 \times 3}, \quad \text{rank}(\mathbf{A}) = 2$$

Compact SVD Components:

$$\mathbf{U}_r = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & -\sqrt{3} \\ 1 & \sqrt{3} \\ 2 & 0 \end{bmatrix}, \quad \mathbf{\Sigma}_r = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{bmatrix}, \quad \mathbf{V}_r = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & -\sqrt{3} \\ 1 & \sqrt{3} \\ 2 & 0 \end{bmatrix}$$

3×2 2×2 3×2

Verification:

$$\mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} = \mathbf{A}$$

3×2 2×2 2×3 3×3

Compact SVD Examples

Example 2 (wide full-rank matrix):

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

2×3

Compact SVD Components:

$$U_r = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \Sigma_r = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix}, \quad V_r = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & 0 \end{bmatrix}$$

2×2 2×2 3×2

Verification:

$$U_r \Sigma_r V_r^\top = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = A$$

$2 \times 2 \quad 2 \times 2 \quad 2 \times 3 \quad 2 \times 3$

Compact SVD Examples

Example 3 (tall full-rank matrix):

$$A = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}_{3 \times 1}$$

Compact SVD Components:

$$U_r = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \Sigma_r = [3], \quad V_r = [1]$$

Verification:

$$\underbrace{U_r}_{3 \times 1} \underbrace{\Sigma_r}_{1 \times 1} \underbrace{V_r^T}_{1 \times 1} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}_{3 \times 1} = A$$

Part V Application of SVD in Data Compression

Sec. 7.1

22 mins

Reading: SVD and Data Compression Using Low-rank Matrix Approximation
<https://dustinstansbury.github.io/theclevermachine/svd-data-compression>

Approximate Compression of Data Matrix

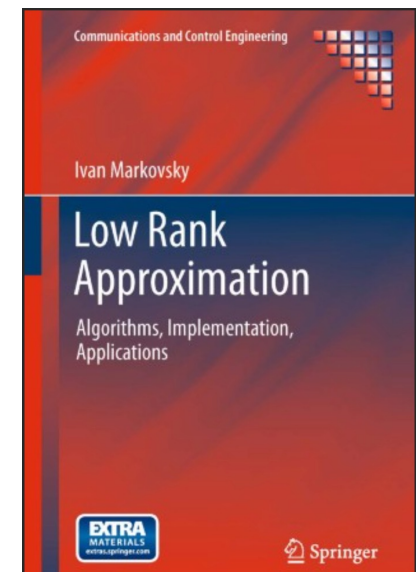
In practice, the rank r is often large, but can
throw away small components as approximation!

Example: $A = \mathbf{u}_1 \mathbf{v}_1^T + 0.01 \mathbf{u}_2 \mathbf{v}_2^T$,
Then $\hat{A} = \mathbf{u}_1 \mathbf{v}_1^T$ is a good approximation of A .

We can make this more concrete:

If this is SVD, then $\|A - \hat{A}\|_F = 0.01$, where

this measures the "approximation error"



Definition (Frobenius norm)

For a matrix $A = (a_{ij})_{m \times n}$, $\|A\|_F \triangleq \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{i,j}^2}$

$$\text{vec} \quad \|v\| = \sqrt{\sum_i v_i^2}$$

Frobenius Norm and Trace

$$\text{Claim 1: } \|B\|_F = \sqrt{\text{Tr}(B^T B)}.$$

Proof of Claim 1 for $n=2$:

$$\begin{aligned}
 B &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} & \|B\|_F^2 &= a^2 + b^2 + c^2 + d^2 \\
 \text{Tr}(B^T B) &= \text{Tr} \left(\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) \\
 &= \text{Tr} \left(\begin{bmatrix} a^2 + c^2 & ab + cd \\ ba + dc & b^2 + d^2 \end{bmatrix} \right) \\
 &= (a^2 + c^2) + (b^2 + d^2)
 \end{aligned}$$

$$\text{Corollary: For a column vector } \mathbf{u}, \text{Tr}(\mathbf{u}\mathbf{u}^T) = \|\mathbf{u}\|^2.$$

Proof: Let $B = \mathbf{u}^T$, then by Claim 1:

$$\text{Tr}(\mathbf{u}\mathbf{u}^T) = \|B\|_F^2 = \|\mathbf{u}^T\|_F^2 = \boxed{u_1^2 + \dots + u_n^2} = \|\mathbf{u}\|^2$$

$\downarrow$ def of $\|\cdot\|_F$ $\downarrow$ def of $\|\cdot\|$

Frobenius Norm: Expressions

Suppose compact SVD of A is $A = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^T$.

$$\textbf{Claim 2: } \|A\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_r^2}.$$

$$\textbf{Corollary: } \|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_r^2}, \text{ where } A_k = \sum_{j=1}^k \sigma_j \mathbf{u}_j \mathbf{v}_j^T.$$

Proof: $A - A_k = \sum_{j=k+1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^T$

Applying Claim 2 directly to $A - A_k$:

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_r^2}$$

e.g. $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \underbrace{\sigma_2 \mathbf{u}_2 \mathbf{v}_2^T}_{A_1}$

$$A_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$$

$$\|A - A_1\|_F = \|\sigma_2 \mathbf{u}_2 \mathbf{v}_2^T\|_F = \sigma_2.$$

Frobenius Norm: Expressions

Claim 2: $\|A\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_r^2}$.

Proof of Claim 2 for $n=2$:

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T$$

$$(u_1 v_1^T)^T = \frac{u_1^T u_1}{v_1 u_1^T}$$

$$\begin{aligned} \|A\|_F^2 &= \text{tr}(A^T A) \\ &= \text{tr}[(\sigma_1 \boxed{v_1 u_1^T} + \sigma_2 \boxed{v_2 u_2^T}) (\sigma_1 \underline{u_1 v_1^T} + \sigma_2 \underline{u_2 v_2^T})] \\ &= \text{tr}(\sigma_1^2 \boxed{v_1 u_1^T \cdot u_1 v_1^T} + \sigma_1 \sigma_2 \boxed{v_1 u_1^T \cdot u_2 v_2^T} + \sigma_2 \sigma_1 \boxed{v_2 u_2^T \cdot u_1 v_1^T} + \sigma_2^2 \boxed{v_2 u_2^T \cdot u_2 v_2^T}) \\ &= \text{tr}(\sigma_1^2 \boxed{v_1 v_1^T}) + \boxed{0} + \boxed{0} + \text{tr}(\sigma_2^2 \boxed{v_2 v_2^T}) \\ &= \sigma_1^2 \text{tr}(\boxed{v_1 v_1^T}) + \sigma_2^2 \text{tr}(\boxed{v_2 v_2^T}) \\ &= \sigma_1^2 \boxed{\|v_1\|^2} + \sigma_2^2 \boxed{\|v_2\|^2} \\ &= \sigma_1^2 + \sigma_2^2 \end{aligned}$$

$$u_i^T u_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}, \quad v_i^T v_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$u_1^T u_1 = u_2^T u_2 = 1$
 $u_2^T u_1 = u_1^T u_2 = 0$

$\Downarrow$
 $\|v_1\|^2 = \|v_2\|^2 = 1$

Frobenius Norm: Expressions

Claim 2: $\|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2}$.

Proof of Claim 2 for general n :

$$\begin{aligned}\|A\|_F^2 &= \text{tr}(A^\top A) = \text{tr}\left(\left(\sum_{i=1}^r \sigma_i \mathbf{v}_i \mathbf{u}_i^\top\right) \left(\sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^\top\right)\right) = \text{tr}\left(\sum_{i,j} \sigma_i \sigma_j \mathbf{v}_i \mathbf{u}_i^\top \mathbf{u}_j \mathbf{v}_j^\top\right) \\ &= \text{tr}\left(\sum_{j=1}^r \sigma_j^2 \mathbf{v}_j \mathbf{v}_j^\top\right) = \sum_{j=1}^r \sigma_j^2 \text{tr}(\mathbf{v}_j \mathbf{v}_j^\top) = \sum_{j=1}^r \sigma_j^2\end{aligned}$$

Taking square root on both sides:

$$\|A\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2}$$

Example of a special matrix

σ_j 's are singular values
(more discussion in Lec 26)

The figure shows distribution of singular values of the

evil Hilbert matrix $H(i, j) = (i + j - 1)^{-1}$

$$H(2,3) = \frac{1}{2+3-1} = \frac{1}{4}.$$

Small singular values only occupy a small portion, thus can be discarded.

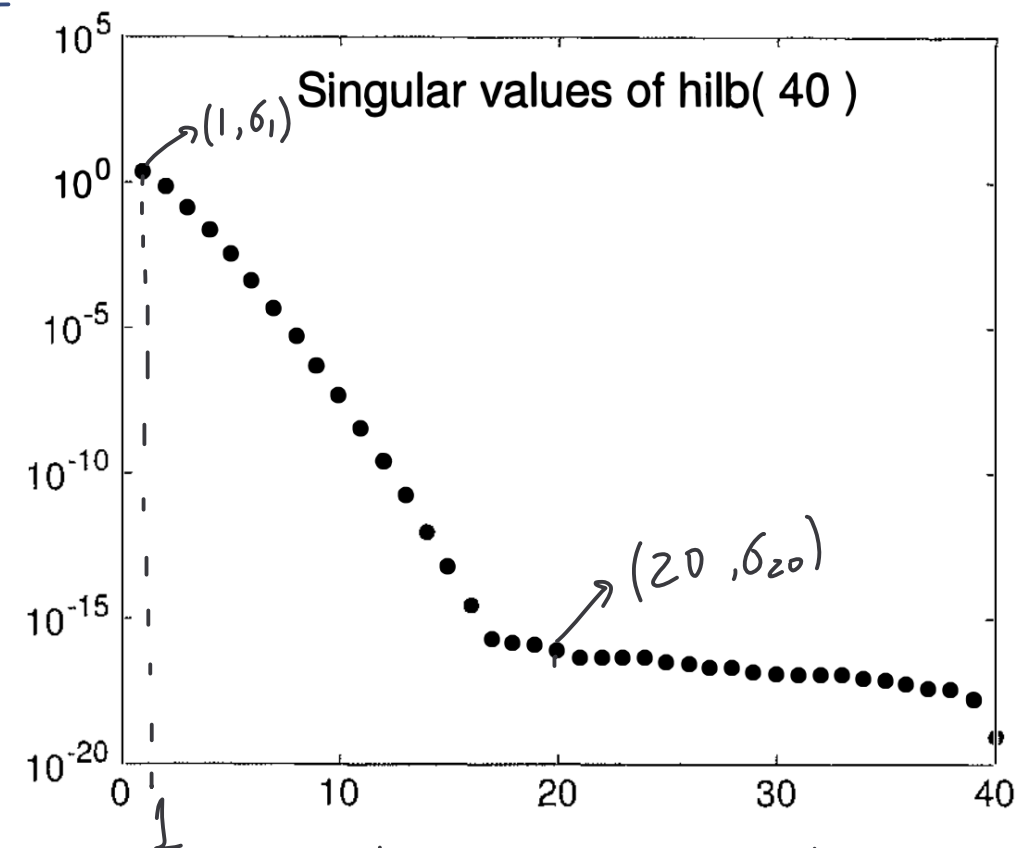
More concrete computation next page.

Compute $\sigma_1, \sigma_2, \dots, \sigma_{40}$

Draw points $(1, \sigma_1), (2, \sigma_2), \dots, (40, \sigma_{40})$

X-axis: index $k=1, 2, \dots, 40$.

$$H = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \dots \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \dots \\ \frac{1}{3} & \frac{1}{4} & \dots \\ \frac{1}{4} & \dots \\ \vdots & \ddots \end{bmatrix}$$



lots of tiny σ_i 's.

Example of a special matrix

This page shows:

For H , using a rank-5 matrix H_5 can approximate well
(lost <0.0003% "energy", i.e. $\|H - H_5\|_F < 0.0003\% \|H\|_F$)

evil Hilbert matrix $H(i, j) = (i + j - 1)^{-1}$

Goal: Approximate H with rank $k = 5$ (keeping top 5 singular values).

1. Compute Frobenius norm of H (total energy):

$$\|H\|_F = \sqrt{\sum_{i=1}^{40} \sigma_i^2} \approx 1.77 \quad (\text{from numerical computation})$$

2. Error of rank-5 approximation (H_5):

$$\|H - H_5\|_F = \sqrt{\sum_{i=6}^{40} \sigma_i^2} \approx 4.37 \times 10^{-6}$$

• Relative error:

$$\frac{\|H - H_5\|_F}{\|H\|_F} \approx 2.47 \times 10^{-6} \quad (0.000247\%)$$

3. Energy retained by rank-5 approximation:

$$\frac{\sum_{i=1}^5 \sigma_i^2}{\sum_{i=1}^{40} \sigma_i^2} \approx 99.9998\%$$

Reading: Python Code for Last Page

python

```
import numpy as np
from scipy.linalg import hilbert

# Generate evil Hilbert matrix
H = hilbert(40)
U, S, V = np.linalg.svd(H)

# Rank-5 approximation
k = 5
H_k = (U[:, :k] * S[:k]) @ V[:k, :]

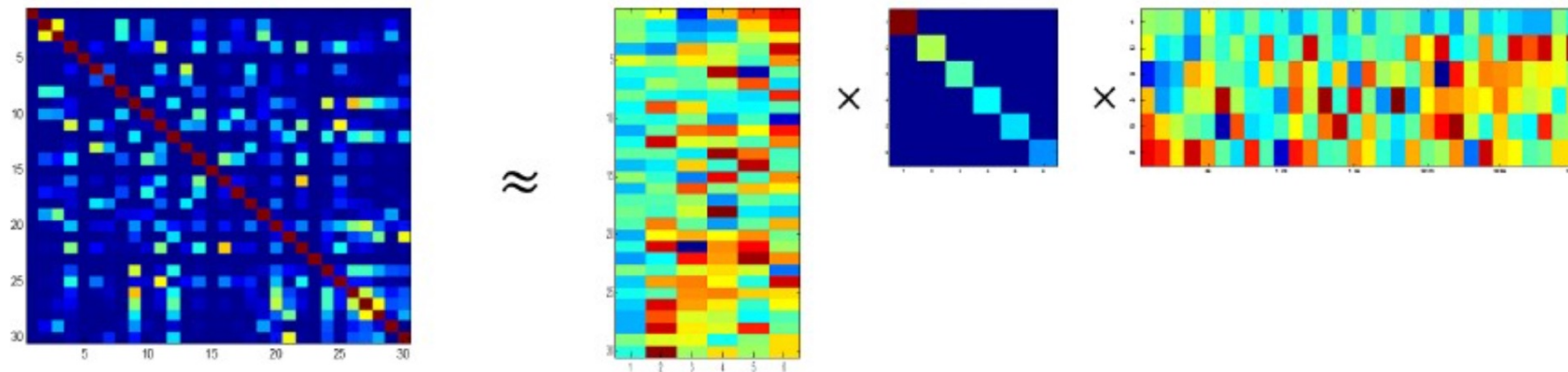
# Compute errors
error = np.linalg.norm(H - H_k, 'fro')
total_norm = np.linalg.norm(H, 'fro')
print(f"Relative error (rank-{k}): {error / total_norm:.3e}")
```

Output:

Relative error (rank-5): 2.47e-06

Compression of Images

Visual quality can be preserved even with a big reduction in the rank.

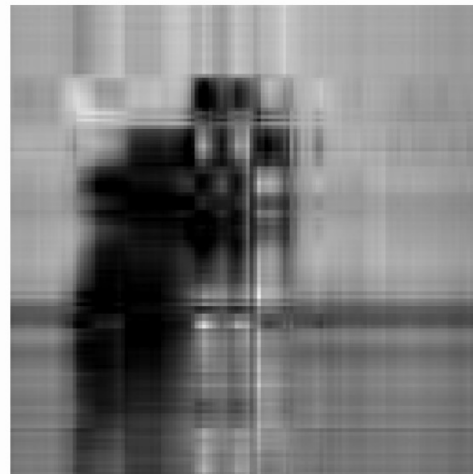


Compression of Images

X



$\hat{X}_4$



Compression: 93.7%
Info. Retained 46.5%

$\hat{X}_{32}$



Compression: 49.8%
Info. Retained 80.4%

Source: SVD and Data Compression Using Low-rank Matrix Approximation
<https://dustinstansbury.github.io/theclevermachine/svd-data-compression>

Low-Rank Approximation: Movie Recommendation

Recommendation systems (personalized)

			
Richard	5	1	4
Mary	?	2	5
Steve	4	3	2

Assumption: **low-rank**.

Question: Can we find SVD of the rating matrix,
from a subset of the entries?

Guaranteed Matrix Completion via
Non-Convex Factorization

Ruoyu Sun and Zhi-Quan Luo, *Fellow, IEEE*

My PhD thesis is
on this problem.



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Instructor's)

One sentence summary:

We learned SVD today.

Detailed summary:

SVD is a decomposition of any real matrix A (can be rectangular) s.t.

$$A = UDV^T, \text{ where}$$

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